The Limits to Growth: Sustainability and the Circular Economy

Lecture 15: Summary and Outlook

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- Updated versions of these slides will be available in our Github repository.

COURSE EVALUATION & FEEDBACK

Course Evaluation / Feedback

W 1617_The Limits to Growth - Sustainability and the Circular Economy (W 1617_WS2526)

TU Clausthal

Rücklauf = 28 Fragebögen

Prof. Dr. Benjamin Leiding

	+	-	Ø
1. Lehrveranstaltung und Lehre / *course and lesson*	0% 20% 40% 60% 80% 100% Skalenbreite: 6		1,17
2. Teilnehmerperspektive / Kompetenzerwerb / Gesamtbewertung /	0% 20% 40% 60% 80% 100% Skalenbreite: 6		1,27

Course Evaluation / Feedback

5.1) Was hat Ihnen an der Lehrveranstaltung besonders gut gefallen ? (Nur innerhalb der Berandung schreiben!)
Which elements of the course did you like in particular? (Please do not write outside the borders.)

- - Entspannte Vortragsweise
  - Vorlesung wird aufgezeichnet und ermöglicht so Lerntempo, Lernzeit und Lernort selbst zu wählen
- Die klare Struktur der Vorlesung und die verständliche Erklärungsweise des Dozenten haben mir sehr geholfen.
- The effort we still have to do, to safe our planet although actions are being taken to solve climate changes problems.
- The Guest Lectures where very interesting, especially when they have experience in the working field.

Course Evaluation / Feedback

5.2) Was hat Ihnen an der Lehrveranstaltung nicht gefallen ? (Nur innerhalb der Berandung schreiben!)
Which parts of the course did you not like? (Please do not write outside the borders.)

- - Vorlesung in Goslar von Clausthal aus nicht sehr gut zu erreichen (Busverbindungen)
  - Abgabepflicht für Übungsblätter erscheint unnötig (Hintergrund unklar)
  - Bei Abgabe der UBs über Cloud-Link gibt es keine Möglichkeit, sich über erfolgreiche Abgabe später zu versichern
- Die Praxisbeispiele und die Übungsaufgaben waren sehr hilfreich, um den theoretischen Stoff besser zu verstehen.
- Some of the topics are a little depressing, but are part of the lecture. Therefore there was nothing bad.

Course Evaluation / Feedback

5.3) Welche Verbesserungsvorschläge haben Sie ? (Nur innerhalb der Berandung schreiben!)
What suggestions for improvement do you have? (Please do not write outside the borders.)

- - Aufgabenabgabe über Moodle
  - Etwas weniger Invited Lectures
- Die interaktive Gestaltung der Lehrveranstaltung und die Möglichkeit, jederzeit Fragen zu stellen.
- The group exercises where very hard to organise.
  Some people just didnt participate, therefore working in a group wasnt possible.

RECAP – LTG COURSE IN A NUTSHELL

LTG Recap

Sustainability – Definition

LTG Recap

Sustainability – Definition

„Development that meets the needs of the present without compromising the ability of future generations to meet their own needs.”

Report of the World Commission on Environment and Development: Our Common Future (1987) – http://www.un-documents.net/our-common-future.pdf

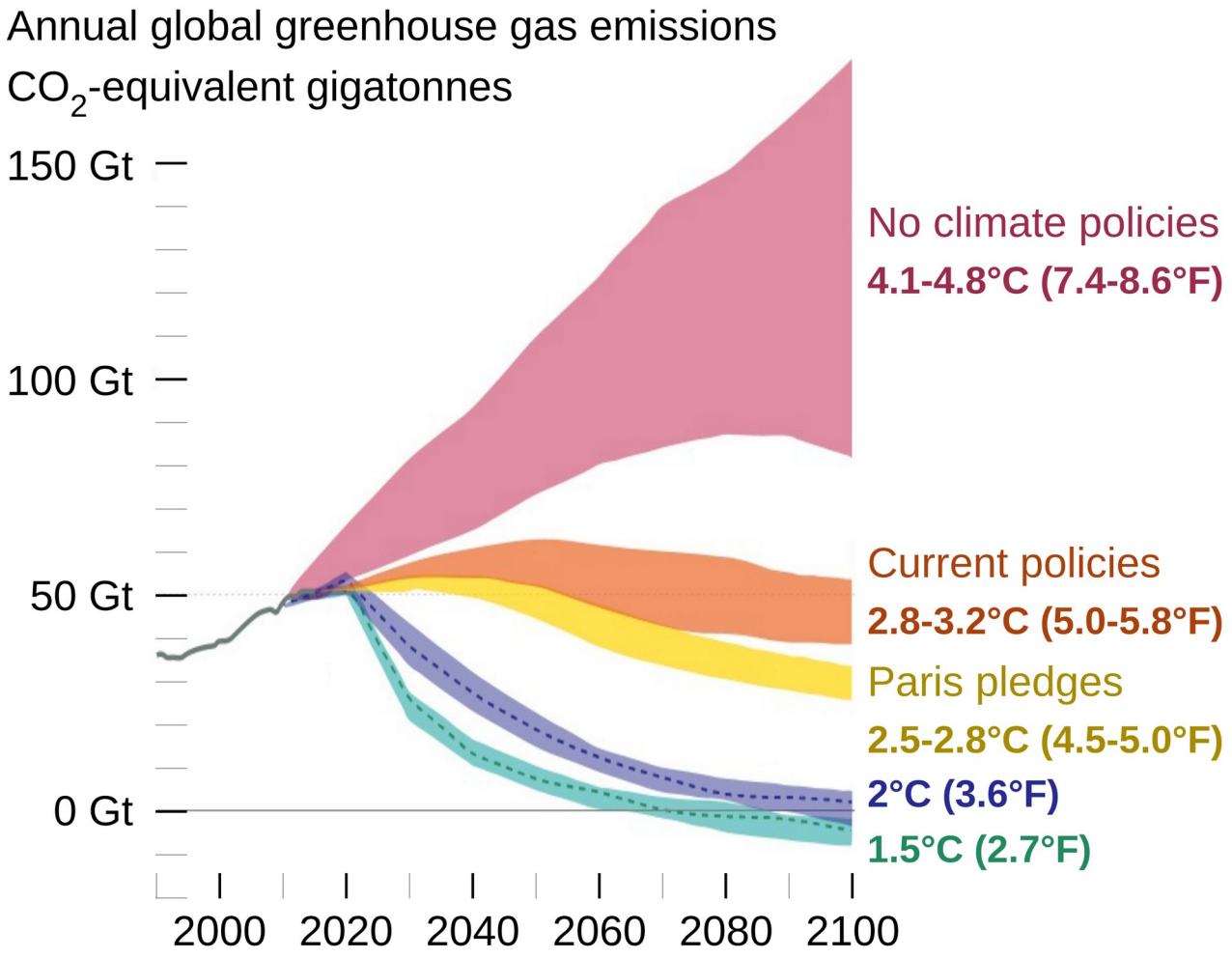
LTG Recap

Sustainability – Implication

Sustainability → Consume less

LTG Recap

Climate Change – Global GHG Emission Pathways (2019)



Hannah Ritchie and Max Roser, adapted for svg and smartphone by Eric Fisk – https://commons.wikimedia.org/wiki/File:Greenhouse_gas_emission_scenarios_01.svg – CC BY-SA 4.0.

LTG Recap

Climate Change – 1.5°C vs. 2/3/4°C

*“If we can keep warming below **3°C** we likely remain within our adaptive capacity as a civilization, but at **2.7°C** warming we would experience great hardship.” -*
Prof. Michael Mann

LTG Recap

Environmental Pollution – Horrible Waste Management





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LTG Recap

Dwindling (Non-)Renewable Resources



"I am afraid of losing my child to a resource war because of a climate collapse"

→ *"Our parents will die of old age, our children will die of climate change"*

XR Strategiepapier 2022 – Pusteblumen und Hype. Bilder: Sebastian Höhn, Joe Pohl, Sandra Doneck, Alessandro Brönnimann

LTG Recap

3 Key Challenges of the 21st Century

1) Climate change / adaption to climate change
2) Environmental pollution
3) Dwindling non-renewable resources

LTG Recap

World3 Standard Run – Business-as-Usual (BAU)

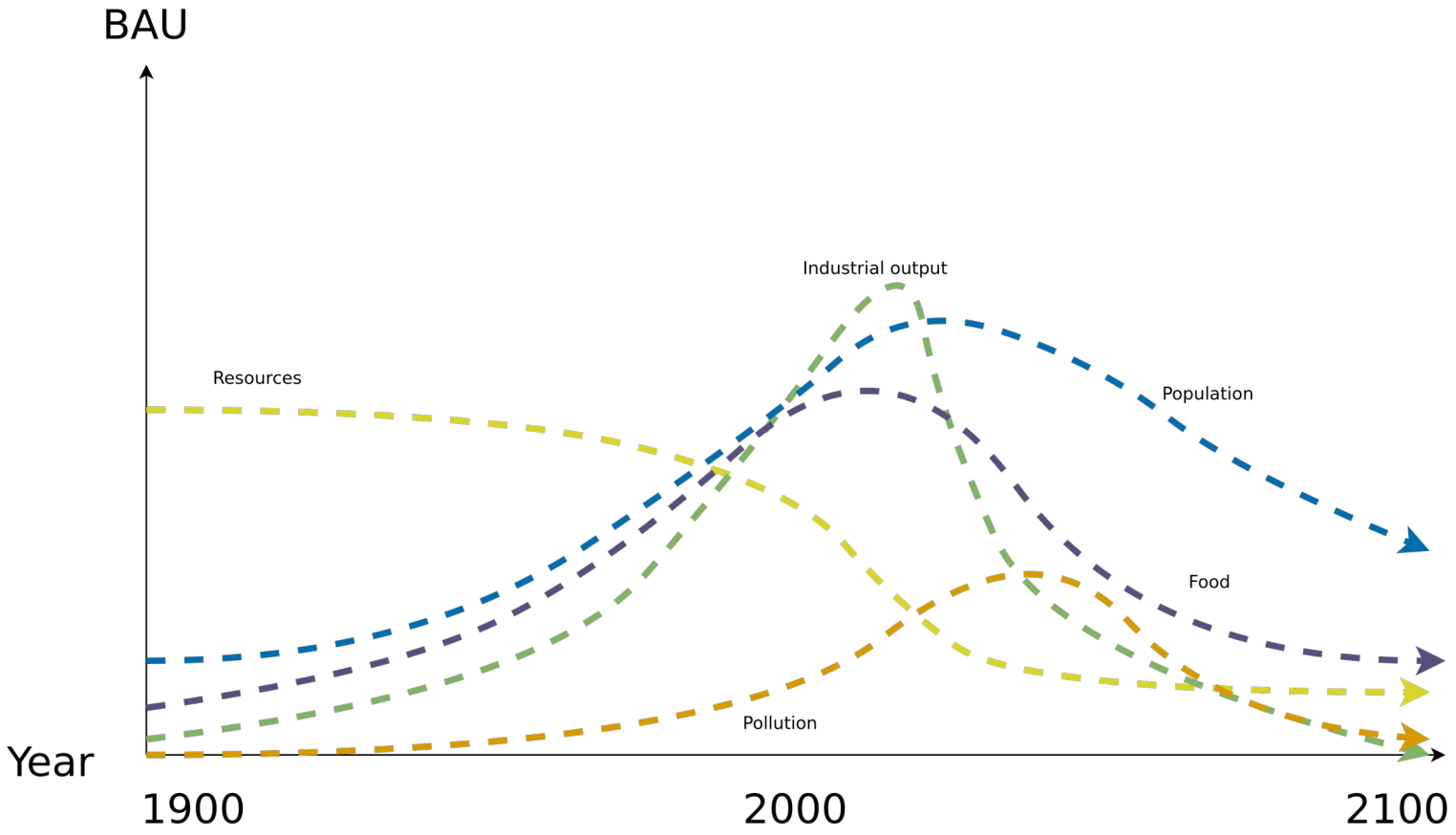
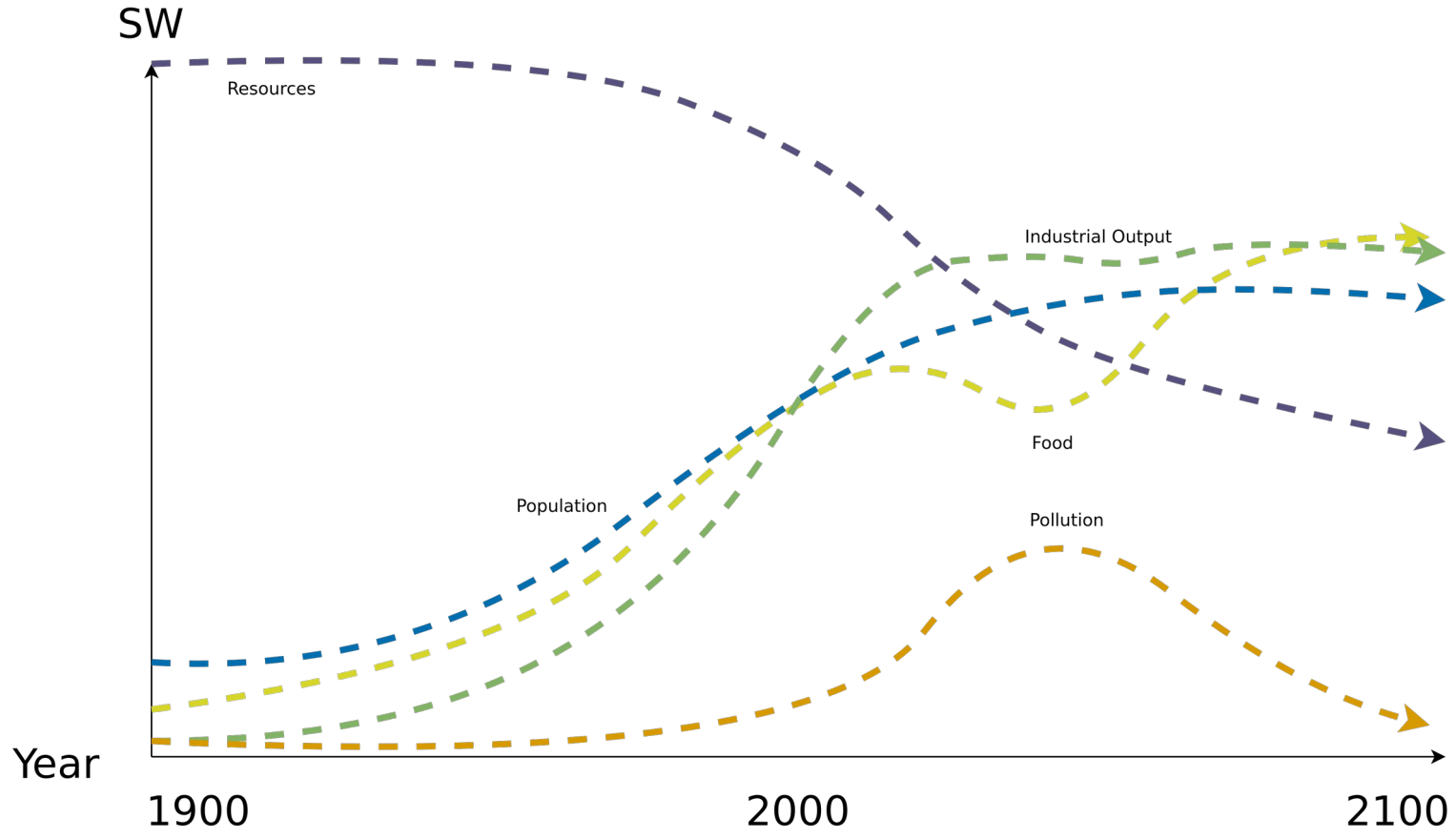


Image recreated from: Gaya Herrington (2020) – Update to limits to growth: Comparing the World3 model with empirical data – Link

LTG Recap

World3 – Stabilized World (SW)

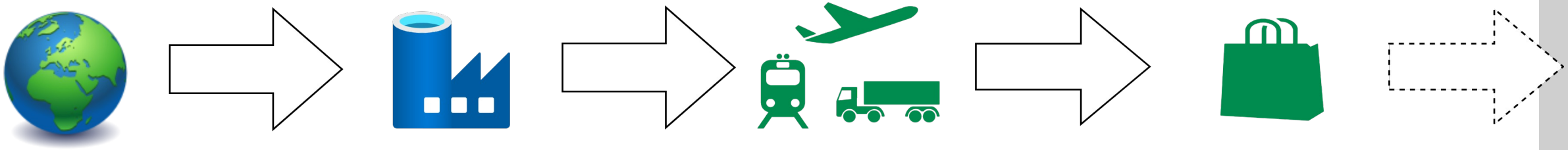


→ Population stabilizes in the twenty-first century, as does human welfare on a high level

Image recreated from: Gaya Herrington (2020) – Update to limits to growth: Comparing the World3 model with empirical data – Link

LTG Recap

Linear (Industrial) Economy



RESOURCES **MANUFACTURING** **DISTRIBUTION** **POINT OF SALE**

GLOBALIZED INDUSTRIALIZED PROCESSES

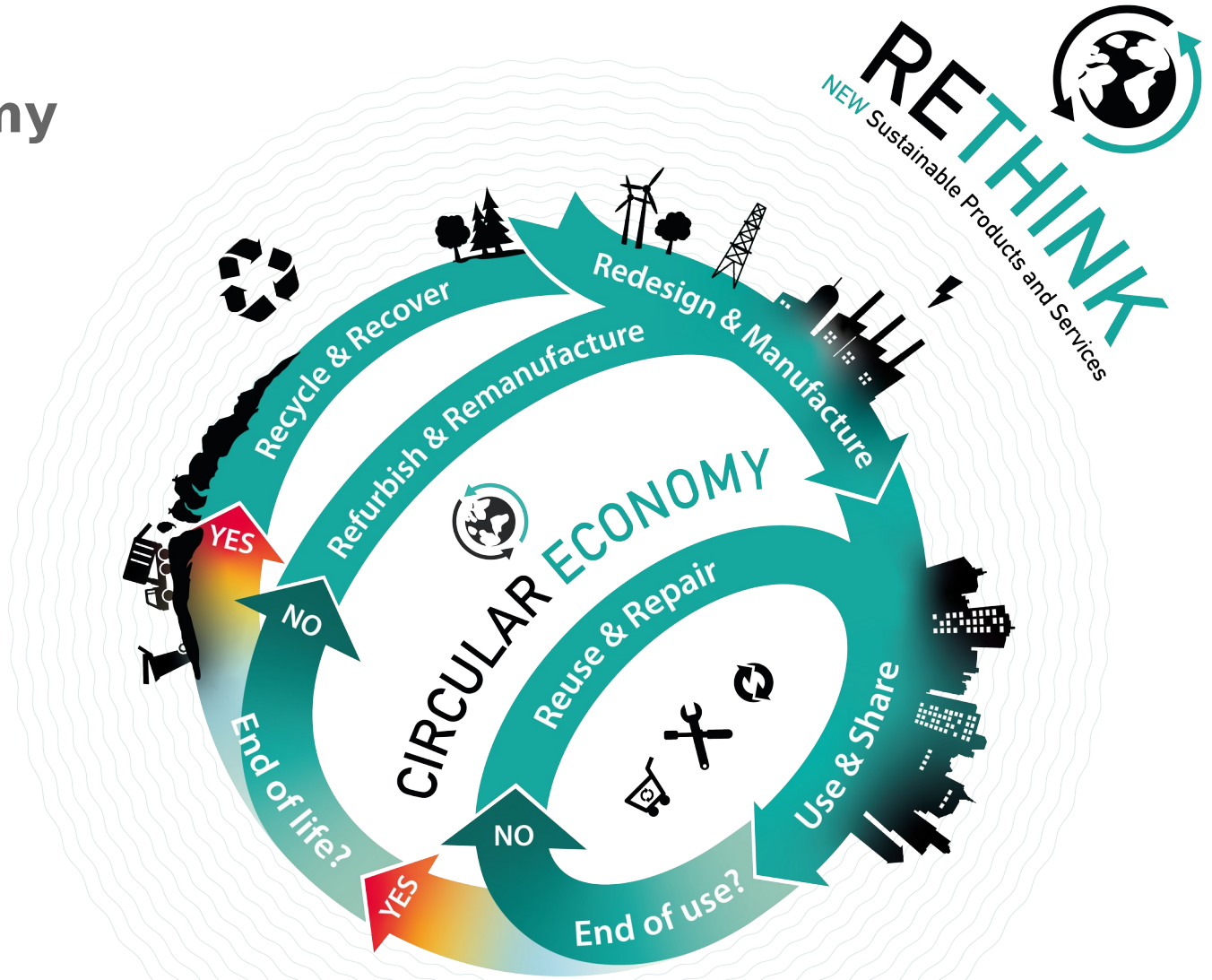
NATURAL RESOURCES (Water, Energy, Materials) and Human Resources (Labor and Skills) are used to produce manufactured **objects** (infrastructure, buildings, equipment, durable projects).

GLOBALIZED DISTRIBUTION LOGISTICS

COMMERCIALIZATION at the factory gate or at point of sale these objects are put to use (sold, leased, rented, sold as service)...

Image adapted from: Walter R. Stahel (2019) – The Circular Economy: A User's Guide.

LTG Recap
Circular Economy



Center for Digital Technologies (DIGIT) (2022) – https://www.digit-research.de/fileadmin/DIGIT/Presse_und_News/Downloads/Grafikpaket_Reallabor_DCE_Kreislaufmodell_V2_ccby.zip – CC BY 4.0.

LTG Recap

Performance Economy – Definition

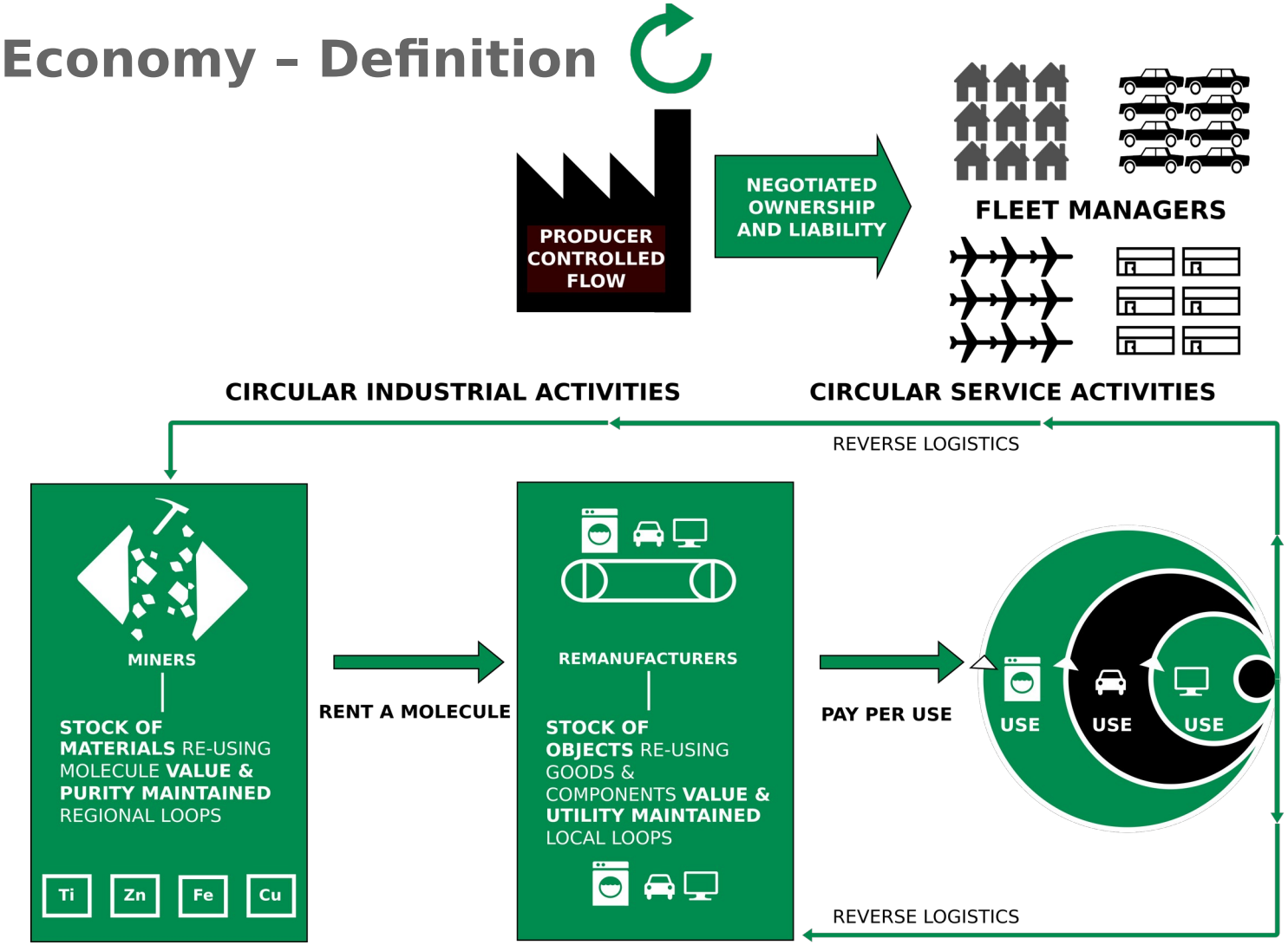


Image adapted from Walter R. Stahel (2019) – The Circular Economy: A User's Guide.

LTG Recap

Circular Society – CE Criticism

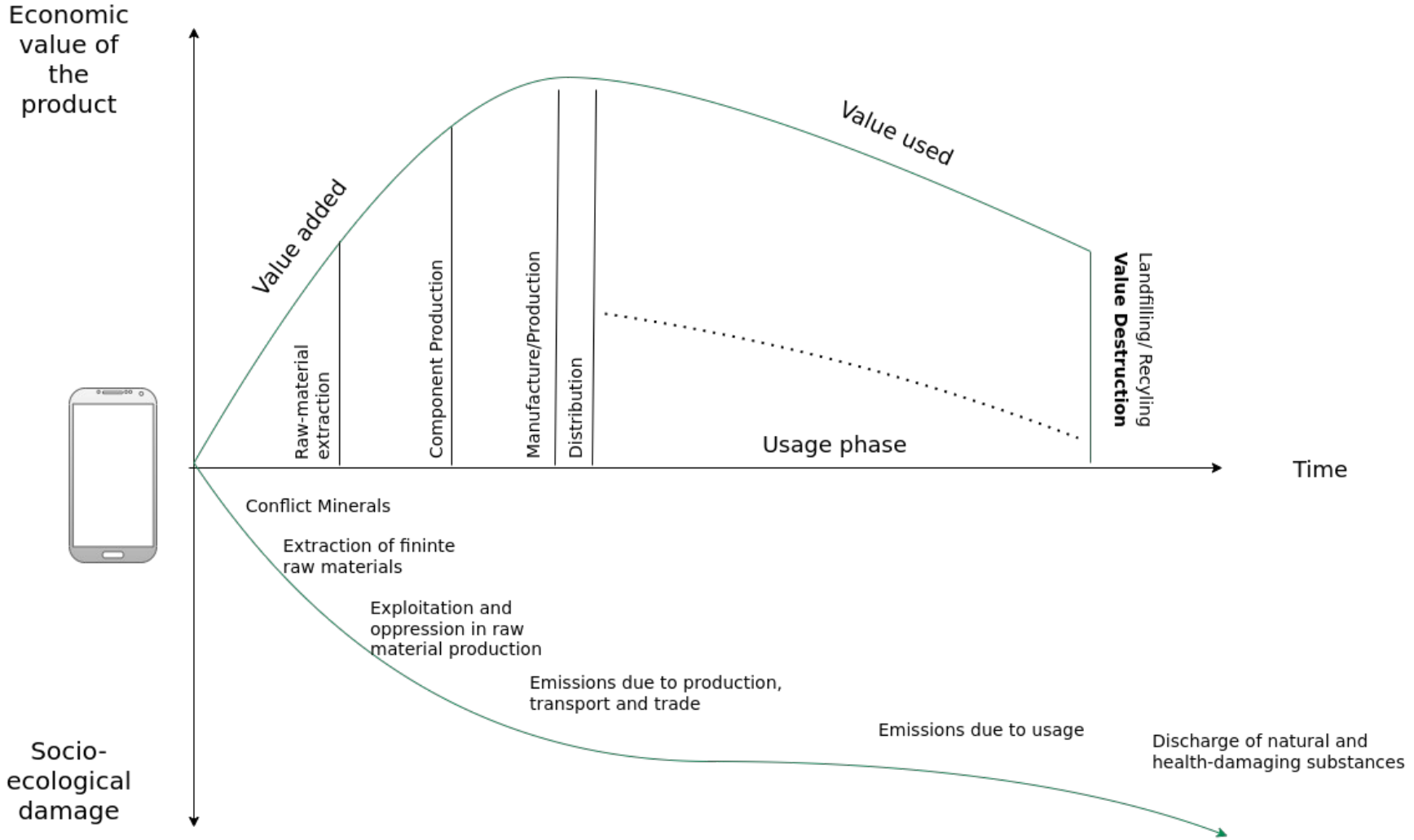


Image adapted from: M. Jaeger-Erben, F. Hofmann (2019) – Kreislaufwirtschaft - Ein Ausweg aus der sozial-ökologischen Krise? – Link.

LTG Recap

Circular Society – CE Criticism

- Replace the LE with circularly oriented forms of consumption and production
- CE focus mostly on earned value management (“Wertschöpfungsmanagement”), product-service systems, product/business model innovations within existing power asymmetries
- Decouple economic growth and consumption of natural resources

→ But why do we need never ending economic growth and why is it good to consume as many goods and services as possible?

Partially based on: F. Hofmann, J. Zwiers (2018) – Circular Society – Eine pluralistische und emanzipatorische Alternative zur Circular Economy? – Link.

LTG Recap

Circular Society – CE Criticism

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- Decouple economic growth and consumption of natural resources

→ But why do we need never ending economic growth and why is it good to consume as many goods and services as possible?

- Alternatives:
  - Sufficiency strategies and lifestyle changes
  - Question the prevailing entrepreneurial orientation towards the shareholder concept
  - Deconstruction of existing power and hegemonic relations

Partially based on: F. Hofmann, J. Zwiers (2018) – Circular Society – Eine pluralistische und emanzipatorische Alternative zur Circular Economy? – Link.

LTG Recap

CS and CE Perspectives

	Holistic	Segmented
Optimist	**Reformist Circular Society** • *Assumptions*: reformed form of capitalism is compatible with sustainability and socio-techincal innovations can enable eco-economic decoupling to prevent ecological collapse. • *Goal:* economic prosperity and human well-being within the biophysical boundaries of the earth. • *Means*: technological breakthroughs, social innovations and new business models that improve ecological health, resource security, and material prosperity for all.	**Techcentric Circular Economy** • *Assumptions:* capitalism is compatible with sustainability and technological innovation can enable eco-economic decoupling to prevent ecological collapse. • *Goal:* sustainable human progress and prosperity without negative environmental externalities. • *Means:* economic innovations, new business models and unprecedented breakthroughs in CE technologies for the closing of resource loops with economic value creation.
Sceptical	**Transformational Circular Society** • *Assumptions:* capitalism is incompatible with sustainability and socio-technical innovation cannot bring absolute eco-economic decoupling to prevent ecological collapse. • *Goal:* a world of conviviality and frugal abundance for all, while fairly distributing the biophysical resources of the earth. • *Means:* complete reconfiguration of the current socio-political system and a shift away from productivist and anthropocentric worldviews to drastically reduce humanity's ecological footprint and ensure that everyone can live meaningfully, and in harmony with the earth.	**Fortress Circular Economy** • *Assumptions:* there is not alternative to capitalism and socio-technical innovation cannot bring absolute eco-economic decoupling to prevent ecological collapse. • *Goal:* maintain geostrategic resource security and earth system stability in global conditions where widespread resource scarcity and human overpopulation cannot provide for all. • *Means:* innovative technologies and business models combined with rationalized resource use, imposed frugality and strict migration and population controls.

Image recreated from: M. C. Friant, R. Salomone, W. J. V. Vermeulen (2020) – A Typology of Circular Economy Discourses: Navigating the Diverse Visions of a Contested Paradigm – Link

LTG Recap

Eco-Sufficiency – Definition(s)

“Seeking enough when more is possible is both intuitive and rational - personally, organizationally and ecologically. And under global ecological constraint, it is ethical.” - Thomas Princen

“A set of policy measures and daily practices that avoid the demand for energy, materials, land, water, and other natural resources while providing wellbeing for all within the planetary boundaries.” - IPCC

Thomas Princen (2005) – The Logic of Sufficiency
IPCC AR6 Synthesis Report: Climate Change 2023 – https://www.ipcc.ch/report/sixth-assessment-report-cycle/

LTG Recap

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Sufficiency → All you need is less

Thomas Princen (2005) – The Logic of Sufficiency
IPCC AR6 Synthesis Report: Climate Change 2023 – https://www.ipcc.ch/report/sixth-assessment-report-cycle/

LTG Recap

Universal Basic Income – Definition(s)

"A transfer that is provided universally, unconditionally, and in cash" – World Bank

"A Basic Income is a periodic cash payment unconditionally delivered to all on an individual basis, without means-test or work requirement." - Basic Income Earth Network

- Periodic → Paid at regular intervals
- Cash payment → Paid in an appropriate medium of exchange
- Individual → Paid on an individual basis (e.g., not to households)
- Universal → Paid to everyone
- Unconditional → Paid without means test and without a requirement to work or to demonstrate willingness-to-work

1.) The World Bank (2020) – Exploring Universal Basic Income : A Guide to Navigating Concepts, Evidence, and Practices - Link
2.) BIEN (Basic Income Earth Network) – https://basicincome.org/about-basic-income

NOW WHAT?

NOW WHAT? → IT'S TIME TO ACT

Act Now

Courage

“But the opposite of hope is not despair. It is grief. Even while resolving to limit the damage, we can mourn. And here, the sheer scale of the problem provides a perverse comfort: we are in this together. The swiftness of the change, its scale and inevitability, binds us into one, broken hearts trapped together under a warming atmosphere.

We need courage, not hope. Grief, after all, is the cost of being alive. [...] ***Courage is the resolve to do well without the assurance of a happy ending.”***

Kate Marvel (2018) – We Need Courage, Not Hope, to Face Climate Change – https://onbeing.org/blog/kate-marvel-we-need-courage-not-hope-to-face-climate-change/

Act Now

“Action is the antidote to despair.”
Joan Baez

Act Now

	Holistic	Segmented
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Act Now

Minimize/Avoid Waste – Waste Is A Human Concept

Waste = Inefficiency

Act Now

Minimize/Avoid Waste – General

- Taking responsibility (being accountable) for the trash/waste that you create and the resources that you consume
  - What is left after you have consumed the product/service?
  - Does it generate, or result in trash/waste?
  - Can you reuse it in any other way, thereby avoiding trash?
  - Is the generated waste, the implicit/explicit environmental pollution, the used resources, and so on, worth the short term pleasure?
- Zero-waste might be unrealistic
- Instead → Reduce your footprint (not only carbon) as much as possible
  - Resources, energy, pollution, emissions, etc.

Act Now

Act Sustainable







Act Now

Act Sustainable





1.Matti Blume – https://upload.wikimedia.org/wikipedia/commons/b/ba/Eviation_Alice%2C_Paris_Air_Show_2019%2C_Le_Bourget_%28SIAE8856%29.jpg – CC BY-SA 4.0
2.Benjamin Leiding – CC BY-SA 4.0.

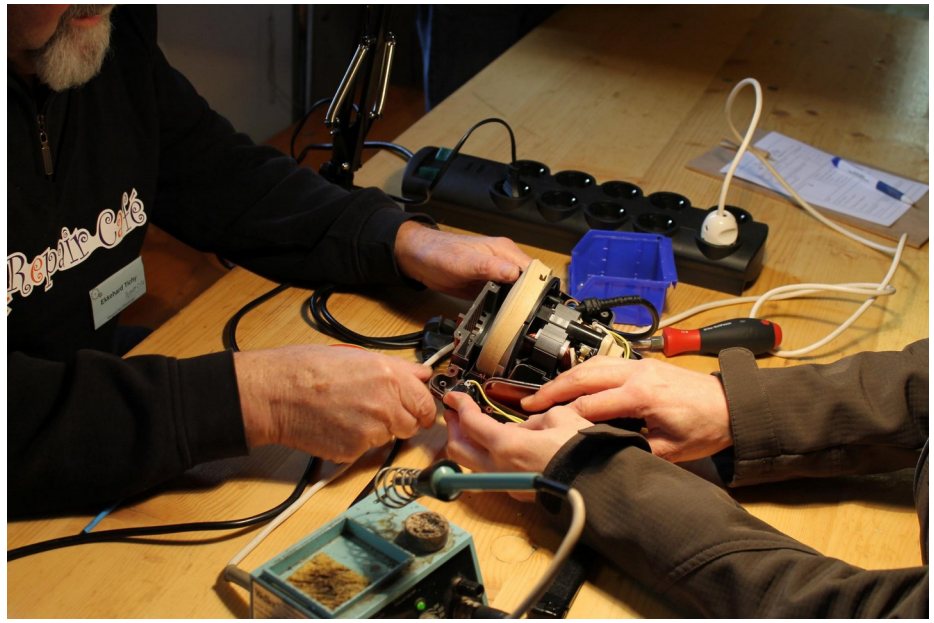
Act Now

Minimize Your Footprint – Sustainability

Sustainability → Consume less

Act Now

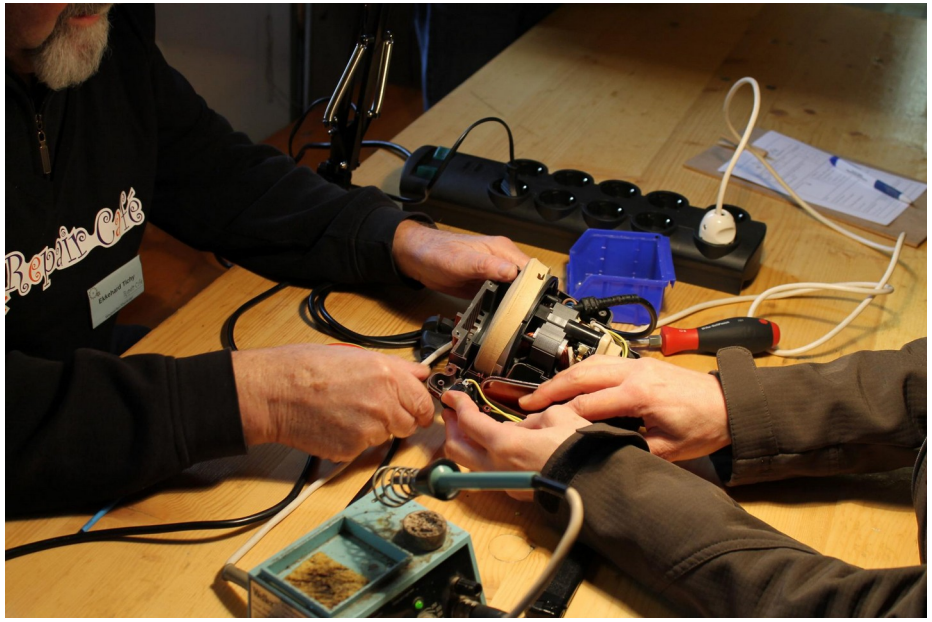
Minimize Your Footprint – Repair Cafe / Sharing / 2nd Hand



1. RaMa2016 – https://upload.wikimedia.org/wikipedia/commons/1/17/Repair_Caf%C3%A9_Trier_Reparatur.jpg – CC BY-SA 4.0
2. Ibrahim Husain Meraj – https://upload.wikimedia.org/wikipedia/commons/9/96/Various_tools_for_cycle_repair_01.jpg – CC BY-SA 4.0
3. Janne Hellsten – https://upload.wikimedia.org/wikipedia/commons/e/e3/Brand_new_second_hand.jpg – CC BY 2.0

Act Now

Minimize Your Footprint – Repair Cafe / Sharing / 2nd Hand

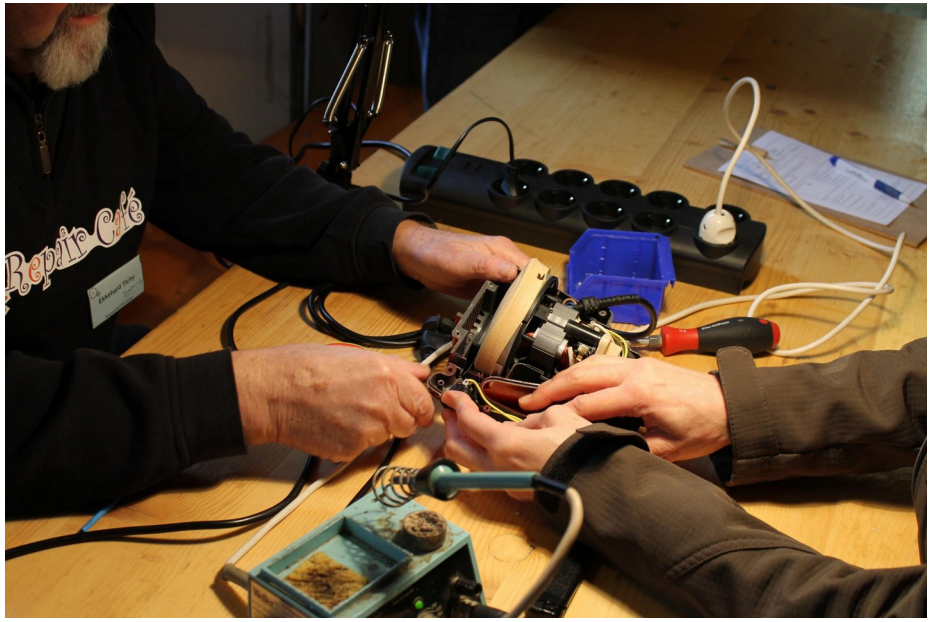




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Act Now

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Act Now

Climate-Resilient Food Production

- Agroforestry → Trees/shrubs are grown around or among crops or pastureland (Link)
- Permaculture → Growth of an agricultural ecosystem in a self-sufficient and sustainable way (Link)
- Silvopasture → Trees, forage and domesticated animals in a mutually beneficial way
- Etc.

Act Now

Climate-Resilient Food Production – Garden Plot (“Kleingarten”)



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Act Now

Climate-Resilient Food Production – Urban/Balcony Gardening







1. Neitram – https://upload.wikimedia.org/wikipedia/commons/7/71/Urban-Gardening-Demonstrationsgarten_der_Hochschule_Augsburg_01.jpg – CC BY-SA 4.0
2. Bad Alley – https://www.flickr.com/photos/badalley/3727226471 – CC BY-NC-SA 2.0
3. Ecoscapes – https://upload.wikimedia.org/wikipedia/commons/9/9d/Condominium_balcony_container_gardening_at_night.jpg – CC BY-SA 4.0

Act Now

Activism – Fridays for Future / Extinction Rebellion / Etc.

- Short history of actions on climate change
  - Geneva, Kyoto, Copenhagen, Paris, Glasgow, etc.
  - Problem
    - → A lot of bla bla
    - → Lack of progress
    - → Policies favor business as usual instead of acting in accordance with science because they are afraid to make unpopular decisions
- **Individual activism as a response / counter movement to inactive governments** who ignore urgent issues (i.e., the three key challenges of the 21st century)

Act Now

Activism – Fridays for Future / Extinction Rebellion / Etc.

“Climate activists are sometimes depicted as dangerous radicals, but the truly dangerous radicals are the countries that are increasing the production of fossil fuels.”

United Nations Secretary General Antonio Guterres

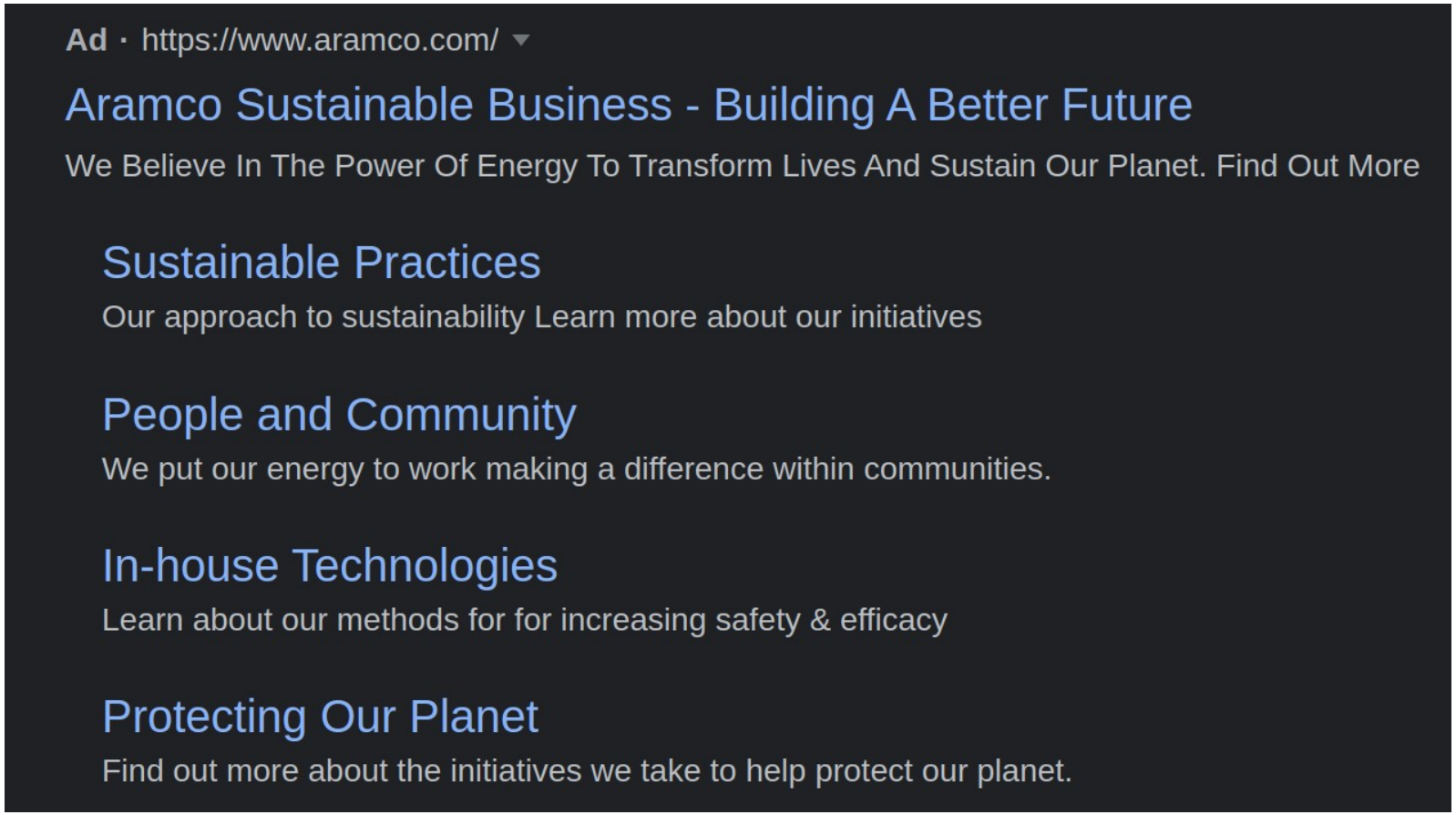
Act Now

Working in a Broken System

- Assumption: Capitalism is a Pyramid/Ponzi scheme that provides disproportionate benefits to few individuals
- What are alternatives?
  - Cooperatives
  - gGmbH (non-profit company with limited liability under German law)
  - Cooperative comunities
- Get a job that does not facilitate an unsustainable use of our planet, i.e., the three key challenges of the 21st century
- Go local (local cooperation, local consumption, local money, local sourcing of resources)
- Less expenses → less time you need to spend working

Fun Fact

Google Search Term “sustainable business cooperative”



Fun Fact

Google Search Term “sustainable business cooperative”



- 100 companies produced more than 70% of the world’s greenhouse gas emissions between 1988 and 2017

- Guess who is on the list?

1) China (Coal) → 14.32%
2) Saudi Arabian Oil Company (Aramco) → 4.50%
3) Gazprom OAO → 3.91%
4) National Iranian Oil → 2.28%
5) ExxonMobil Corp → 1.98%
6) Coal India → 1.87%
7) Petroleos Mexicanos (Pemex) → 1.87%
8) Russia (Coal) → 1.86%
9) Royal Dutch Shell PLC → 1.67%
10) China National Petroleum Corp (CNPC) → 1.56%
11) BP PLC → 1.53%
12) Chevron Corp → 1.31%

Blaming individuals and denying any responsibility → great strategy!

https://www.theguardian.com/sustainable-business/2017/jul/10/100-fossil-fuel-companies-investors-responsible-71-global-emissions-cdp-study-climate-change

Act Now

Wales “One Planet Development”

- “One Wales: One Planet”
  - Timeframe → one generation
  - Goal
    - Wales should use only its fair share of the earth’s resources
    - Zero carbon emissions for both use and construction
- How?
  - Low Impact Development principles (applicable in Wales)
  - Prototypes / Test Communities
    - Single homes
    - Co-operative communities

http://www.oneplanetcouncil.org.uk/

WRAP-UP

Wrap-Up

- Kate Marvel: “*Courage is the resolve to do well without the assurance of a happy ending.*”
  - You do not have to be a sustainability saint → be as sustainable as possible, it still makes a difference
  - Small progress vs. no progress
  - Silence/being inactive supports the destruction of our planet (← I just checked, still no “*planet B*” available)
  - Do something instead of giving up right away
- Leading by example

Thanks!